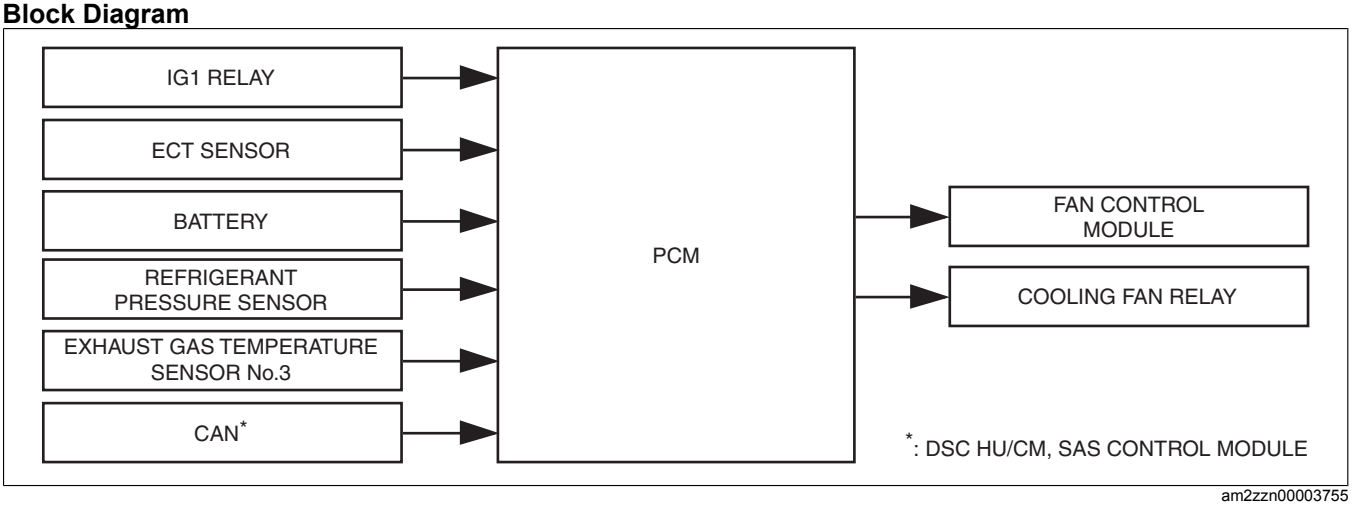
ELECTRICAL FAN CONTROL [SKYACTIV-D 1.5]

id0140z5205000

Outline

- Through cooling of the radiator and condenser by operation of the cooling fan according to vehicle conditions, engine reliability and cooling performance have been improved.
- The PCM determines the engine operation conditions based on the signal from each sensor and performs duty-control of the fan control module to control the rotation speed of the cooling fan motor.

Block Diagram

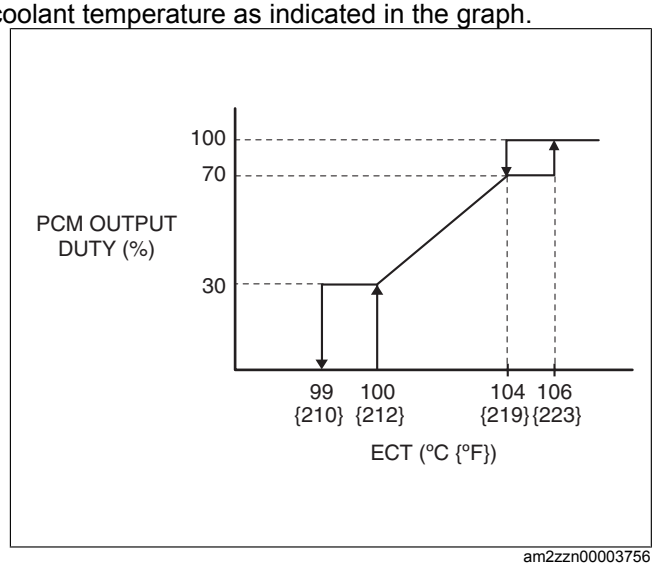


am2zzn00003755

Operation

Engine coolant temperature conditions

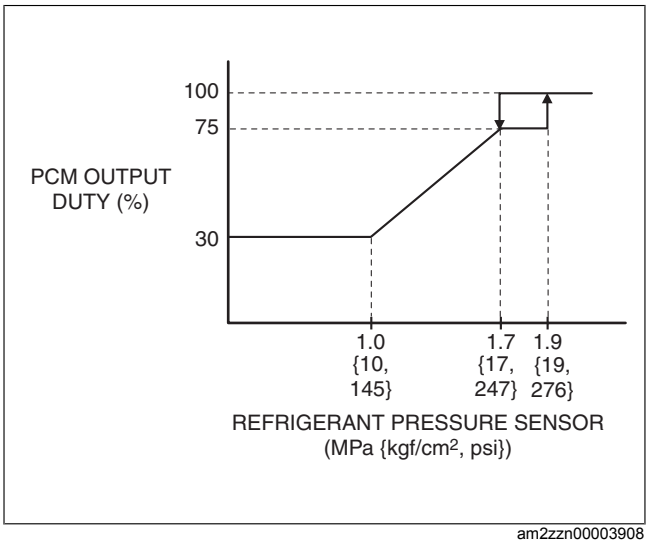
- Output duty ratio is controlled according to the engine coolant temperature as indicated in the graph.
- While the engine coolant temperature is decreasing, it is controlled at a different temperature from that of when the engine coolant temperature is increasing.



am2zzn00003756

A/C conditions

- Output duty ratio is controlled according to the refrigerant pressure when the A/C switch is on.



am2zzn00003908

Exhaust air temperature condition

- If all of the following conditions are met to suppress temperature increase inside the engine compartment, the output duty is controlled at a ratio of 75%.
  - — Exhaust gas temperature sensor No.3: 400 °C {221 °F} or more
  - — Vehicle speed: 40 km/h {25 mph} or less

After cooling control

- To suppress temperature increase inside the engine compartment, the cooling fan operates (output duty ratio: 50 %) for up to 9 min under the following condition.
  - — Exhaust gas temperature sensor No.3: 400 °C {221 °F} or more

Condition during collision

- If a collision signal from the SAS control module is received, the PCM stops the engine cooling fan to avoid use of battery voltage.